Sia R_q: 10101

Calcolare ndcg

Il vettore Gain è $G = (1,0,1,0,1)$

$$DCG[i] = \begin{cases} G(i) & \text{se } i=1 \\ \frac{G(i)}{\log_2 i} + DCG(i-1) & \text{se } i > 1 \end{cases}$$

$DCG = (1, 1, \frac{1}{\log_2 3}+1, \frac{1}{\log_2 3}+1, \frac{1}{\log_2 5}+\frac{1}{\log_2 3}+1)$

$IG = (1,1,1,0,0)$

$IDCG = (1, \frac{1}{\log_2 2}+1, \frac{1}{\log_2 3}+2, \frac{1}{\log_2 3}+2, \frac{1}{\log_2 3}+2)$ — i^2

$nDCG = \left(1, 1/2, \frac{\frac{1}{\log_2 3}+1}{\frac{1}{\log_2 3}+2}, \frac{\frac{1}{\log_2 3}+1}{1/\log_2 3+2}, \frac{\frac{1}{\log_2 5}+\frac{1}{\log_2 3}+1}{\frac{1}{\log_2 3}+2}\right)$ → $nDCG[i] = \frac{DCG[i]}{IDCG[i]}$

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Sia R_q: D3 D1 D5 D7 D9 D8

Rilevanza 3 0 0 0 2 3

$G = (3,0,0,0,2,3)$

$DCG = (3,3,3,3, \frac{2}{\log_2 5}+3, \frac{3}{\log_2 6}+\frac{2}{\log_2 5}+3)$